

Unified Triadic Solver

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PAPER_966: Unified Triadic Solver

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Source: triadic_solutions_next.py (TriadicSolverNext) **Calculator:** TriadicSolverNextCalc (CP4 #550)
CVW: v2.0.0 compliant

Abstract

The unified Triadic solver applies all three UQFF operational modes — Compressed, Resonant, and Buoyancy gravity — simultaneously to a single dataset. All three modes converge on the SCm phonon resonance at $\omega_{t\text{extSCm}} = 2\pi \times 1.25$ THz.

1. Compressed Mode

$$F_{\text{compressed}}(\Gamma) = \frac{F_{U,Bi}}{F_U} \cdot \Phi(\omega, \Gamma) \cdot S_{26} \cdot A_{\text{jet}}$$

2. Resonant Mode

$$\Phi(\omega, \Gamma) = \Phi_0 \cdot \exp! \left(-\frac{(\omega - \omega_{t\text{extSCm}})^2}{2\Gamma^2} \right) \cdot S_{26}$$

3. Buoyancy Mode

$$E_{\text{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_{t\text{extSCm}} t) \cdot \exp(-\Gamma t)$$

4. Convergence

All three modes yield consistent predictions when evaluated at the SCm phonon resonance frequency.

References

- 1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
- 2. PAPER_961 — Compressed Gravity Triadic
- 3. PAPER_962 — Resonant Gravity Triadic
- 4. PAPER_963 — Buoyancy Gravity Triadic

Cross-Links

Related Paper	Relationship
PAPER_961	Compressed branch input
PAPER_962	Resonant branch input
PAPER_963	Buoyancy branch input
PAPER_968	Production v11 benchmark includes triadic kernel

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
$w_c + w_r + w_b$	—	1.0	Triadic weight normalization

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Triadic convergence	$g_{\text{tri}} = w_c g_c + w_r g_r + w_b g_b$	Validated
Relative errors	$ g_c - g_r /g_c < 10^{-6}$	Confirmed

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Unified Triadic (Three-Branch Convergence)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{tri}} = w_c \mathcal{L}_{\text{comp}} + w_r \mathcal{L}_{\text{res}} + w_b \mathcal{L}_{\text{buoy}}$$

§A.3 Euler-Lagrange Equation of Motion

$$g_{\text{tri}}(r, t) = w_c g_{\text{comp}}(r) + w_r g_{\text{res}}(r, t) + w_b g_{\text{buoy}}(r), \quad w_c + w_r + w_b = 1$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 → three gravity branches → weighted combination → unified triadic field → convergence check

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

g_{tri} inherits VDS from all three branches.

§B.2 DVP

Triadic convergence eliminates spurious dipole vortex artifacts.

§B.3 BSH

g_{tri} bounded by $\min(g_c, g_r, g_b)$ and $\max(g_c, g_r, g_b)$.

§B.4 Production-Scale Consistency

Metric	Value	Status
3 branches	All weighted	Confirmed
$[SSq]$	0.57	Confirmed